Welcome to Inverse Circular Functions: Part 5

This extension stage covers the graphs, domains, ranges, and derivatives of arcsec, arccosec, and arccot, together with compositions and identities involving these functions.

How to Navigate:

- Use the **arrow keys** on your keyboard to move between pages
- Use **swiping** gestures on touch devices
- Hover near the **bottom of the page** to reveal navigation buttons

These extension functions build directly on arcsin, arccos, and arctan — and the same implicit differentiation technique from Part 3 applies to their derivatives!

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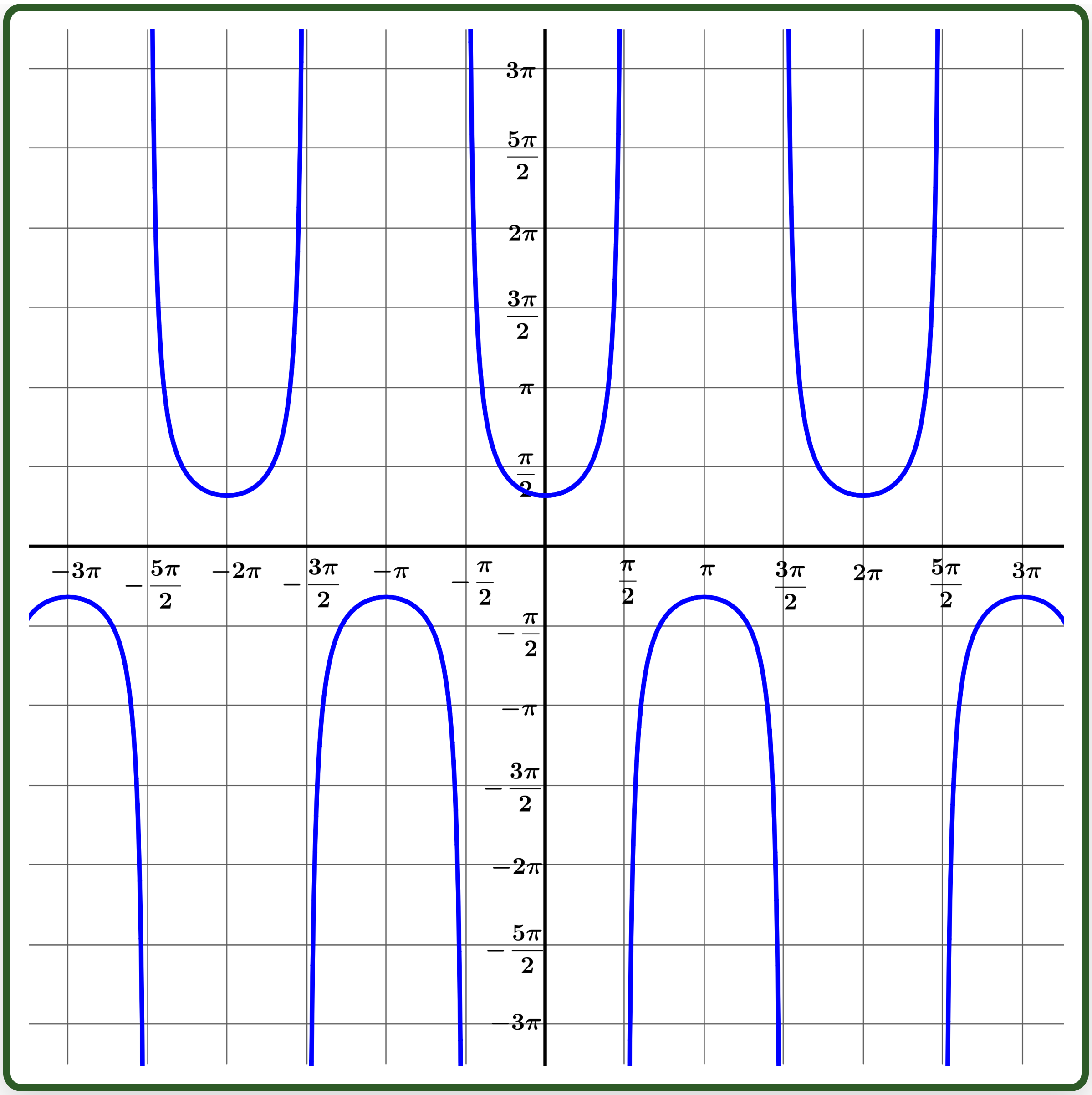
We've worked through the basic circular functions: sin, cos, and tan. We've worked through the reciprocal circular functions: cosec, sec, and cot. We've worked on the inverses of the basic functions: arcsin, arccos, and arctan. Surely that's plenty? Yes it is, especially when we've worked through them all in such depth. But if you are still just a little bit interested to see more, we could work on the inverses of the reciprocal functions: arccosec, arcsec, arccot. Strictly for interest only, though!

Here is the graph $y = \sec x$. Draw this graph, and, on the same set of axes, draw the graph $x = \sec y$.



3π
$\frac{5\pi}{2}$
2π
$\frac{3\pi}{2}$
π
$\frac{\pi}{2}$
-3π
$-\frac{5\pi}{2}$
-2π
$-\frac{3\pi}{2}$
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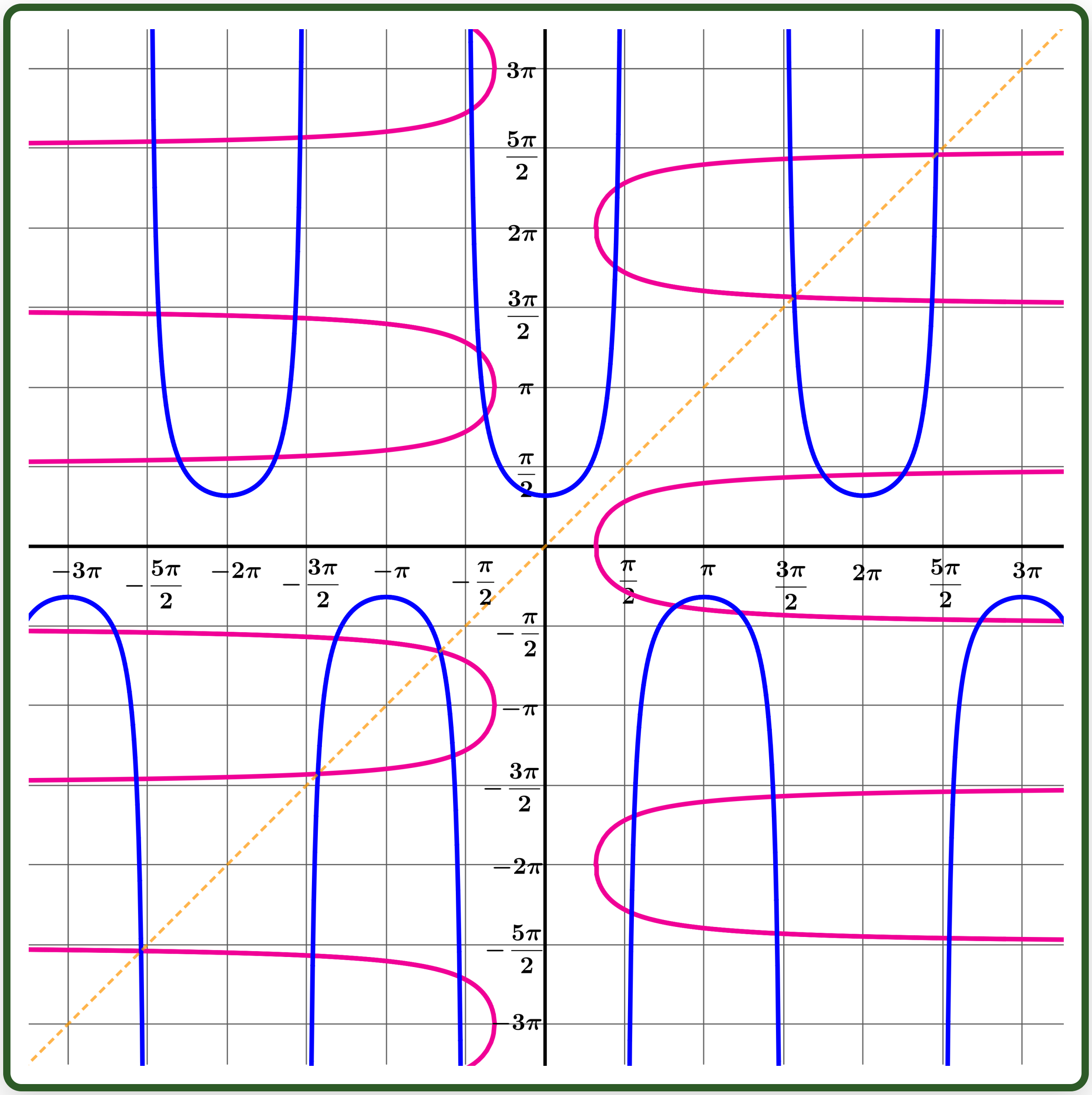

3π
$\frac{5\pi}{2}$
2π
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π
$\frac{\pi}{2}$
-3π
$-\frac{5\pi}{2}$
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$-\pi$
$-\frac{3\pi}{2}$
-2π
$-\frac{5\pi}{2}$
-3π

Why is the graph $x = \sec y$ not the same as the graph $y = \operatorname{arcsec} x$?

[Animation: vertical line test on sec graph — see interactive version]

For the graph to represent a function, the vertical line must only ever intersect the graph at a single point.

How can we choose part of the graph $x = \sec y$ to create a sensible graph of the function $y = \operatorname{arcsec} x$?

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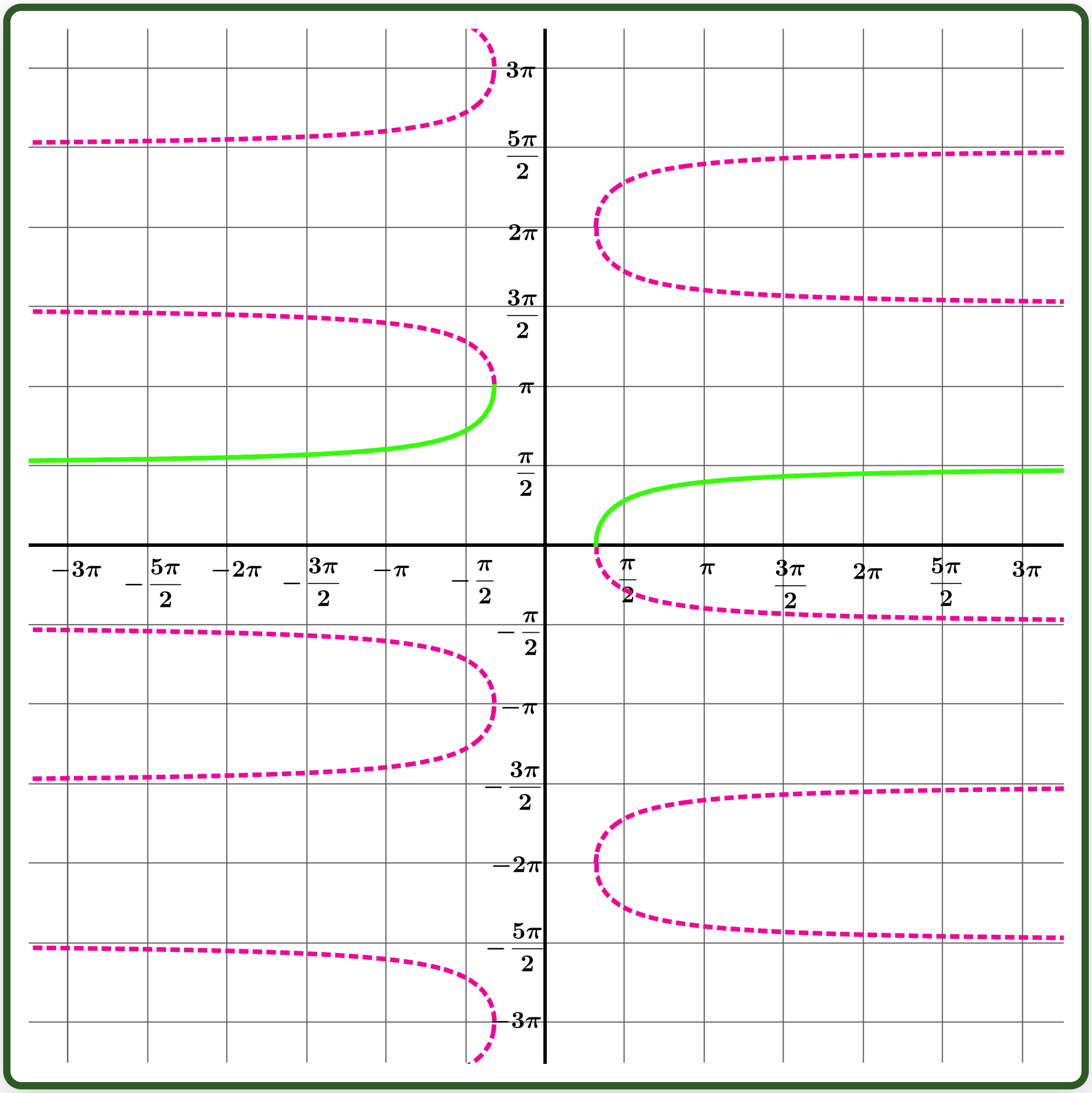
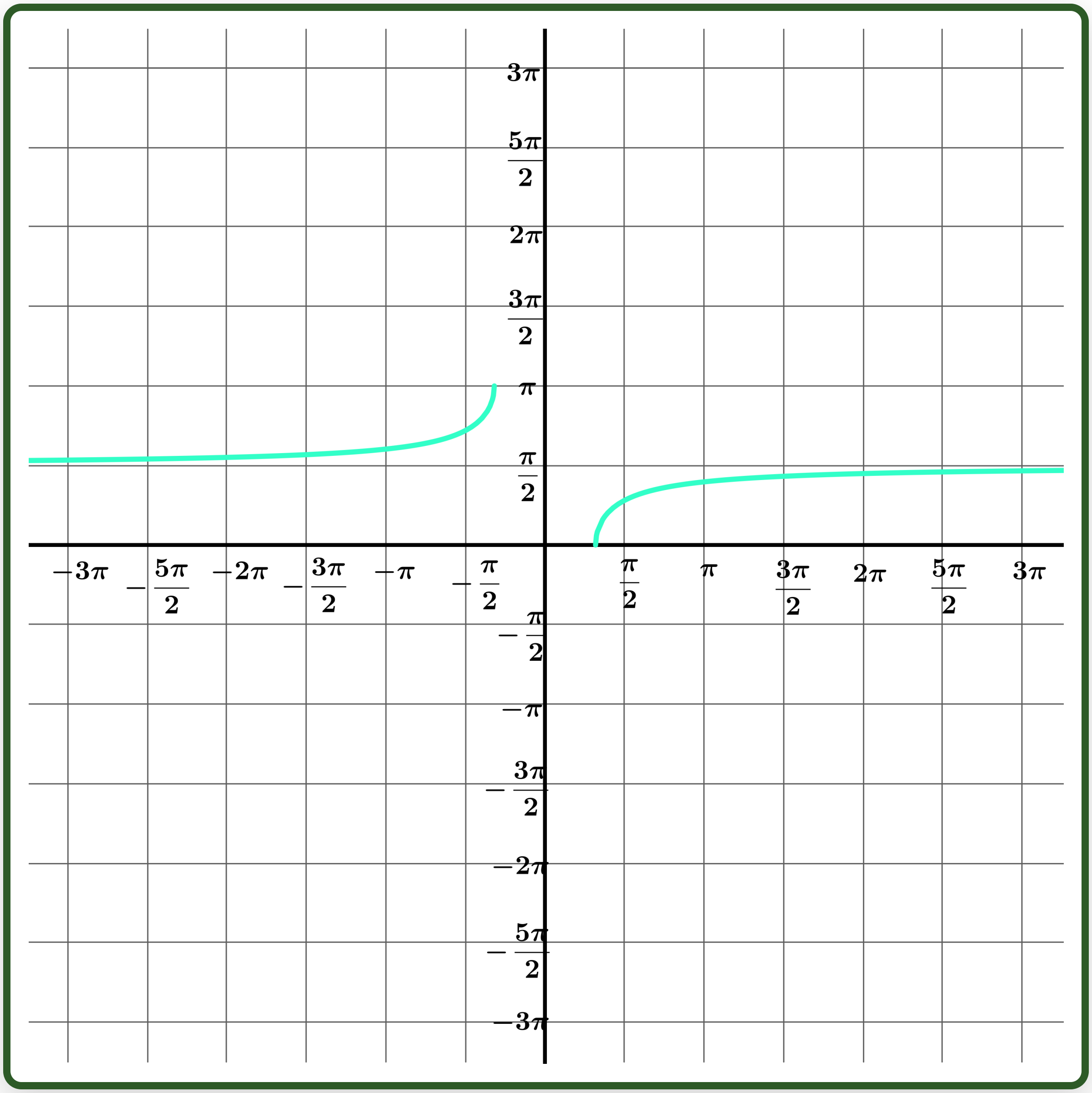


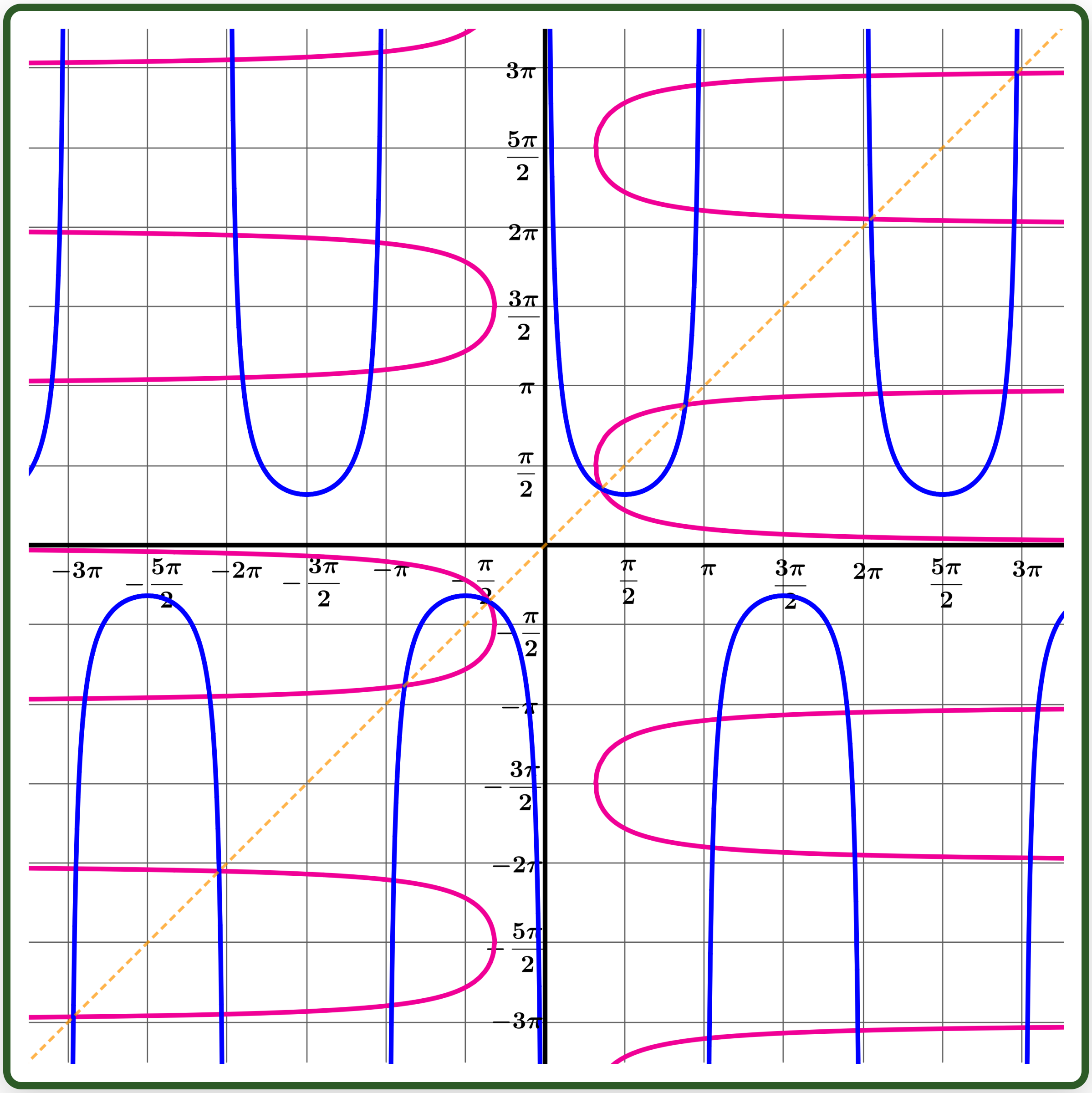
If this is the graph $y = \mathbf{arcsec}\, x$, what are the domain and the range of the function arcsec?



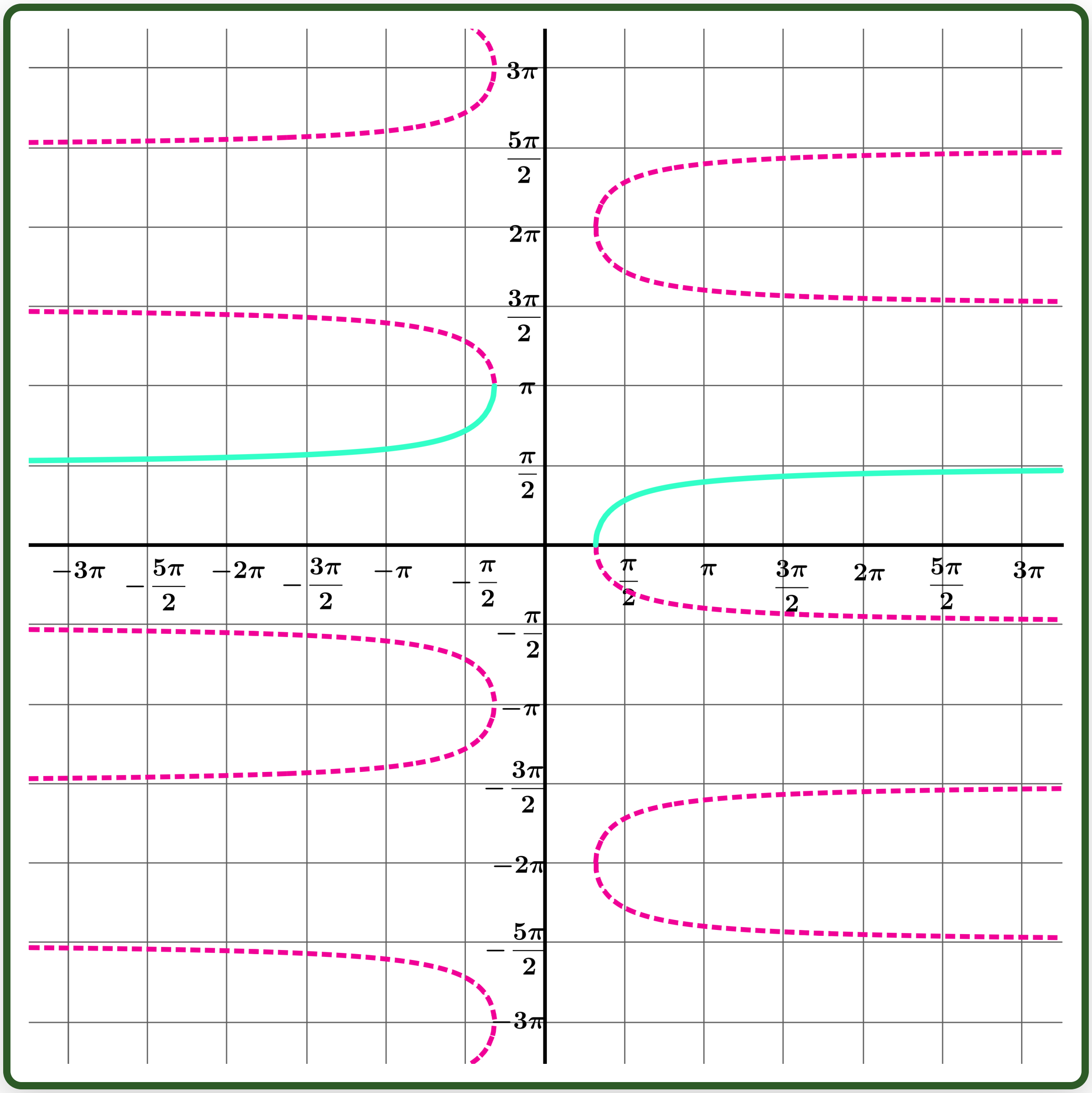
Domain: $(-\infty, -1] \cup [1, \infty)$

Range: $[0,\ \pi] \setminus \{0\}$

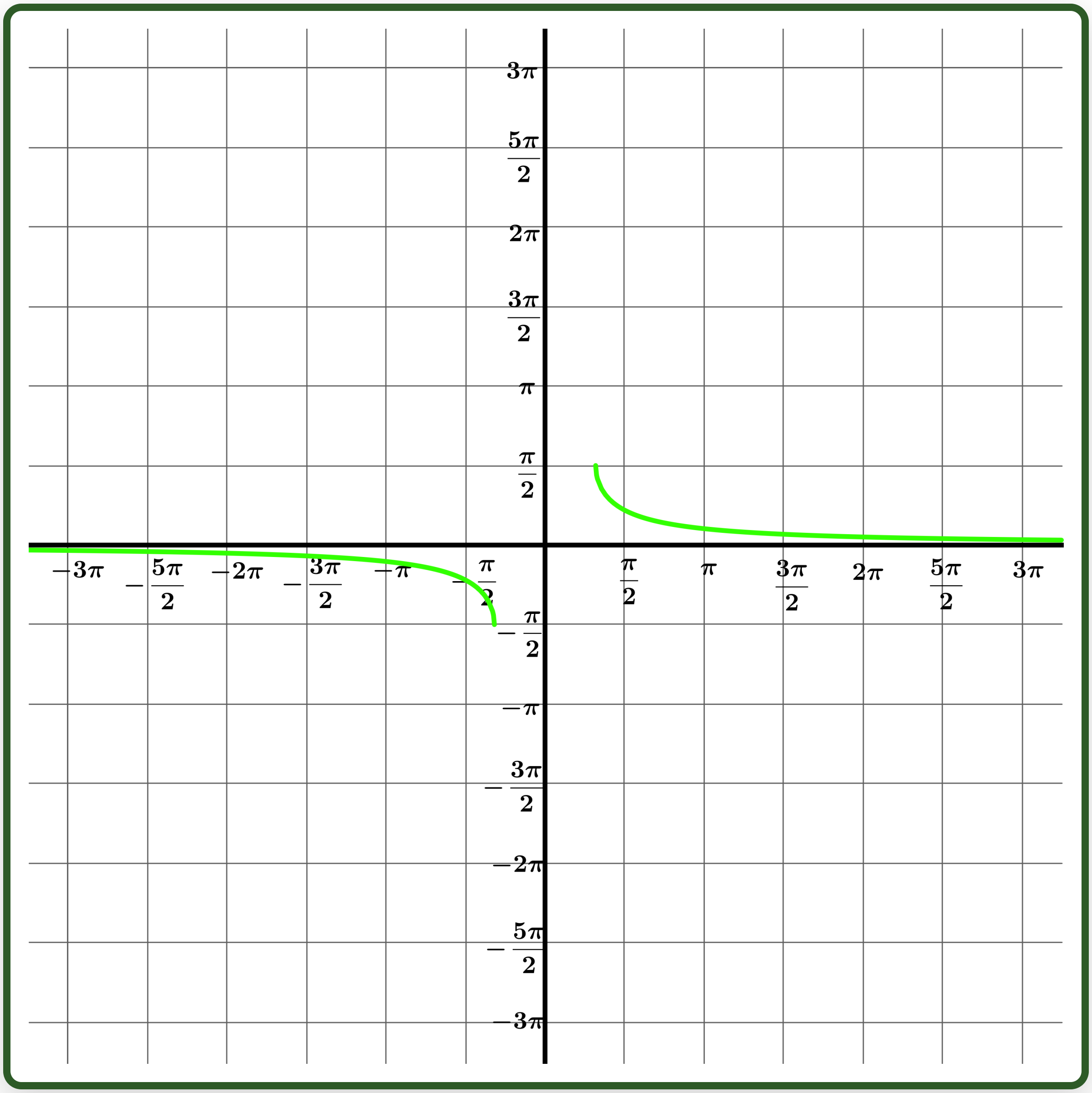
Draw the graphs $y = \operatorname{cosec} x$ and $x = \operatorname{cosec} y$ on the same set of axes.



How can we choose part of the graph $x = \operatorname{cosec} y$ to create a sensible graph of the function $y = \operatorname{arccosec} x$?



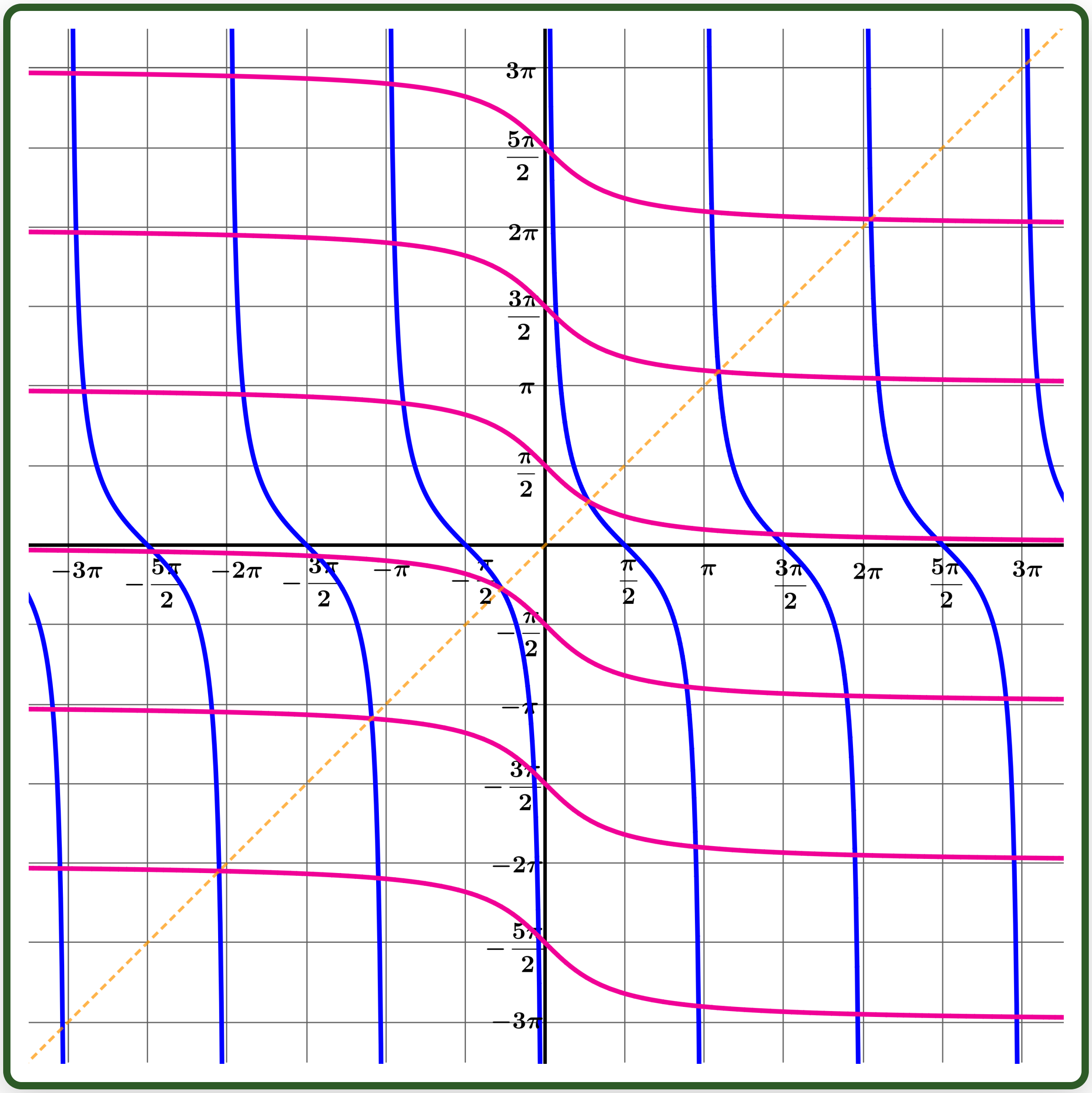
What are the domain and range of the function arccosec?



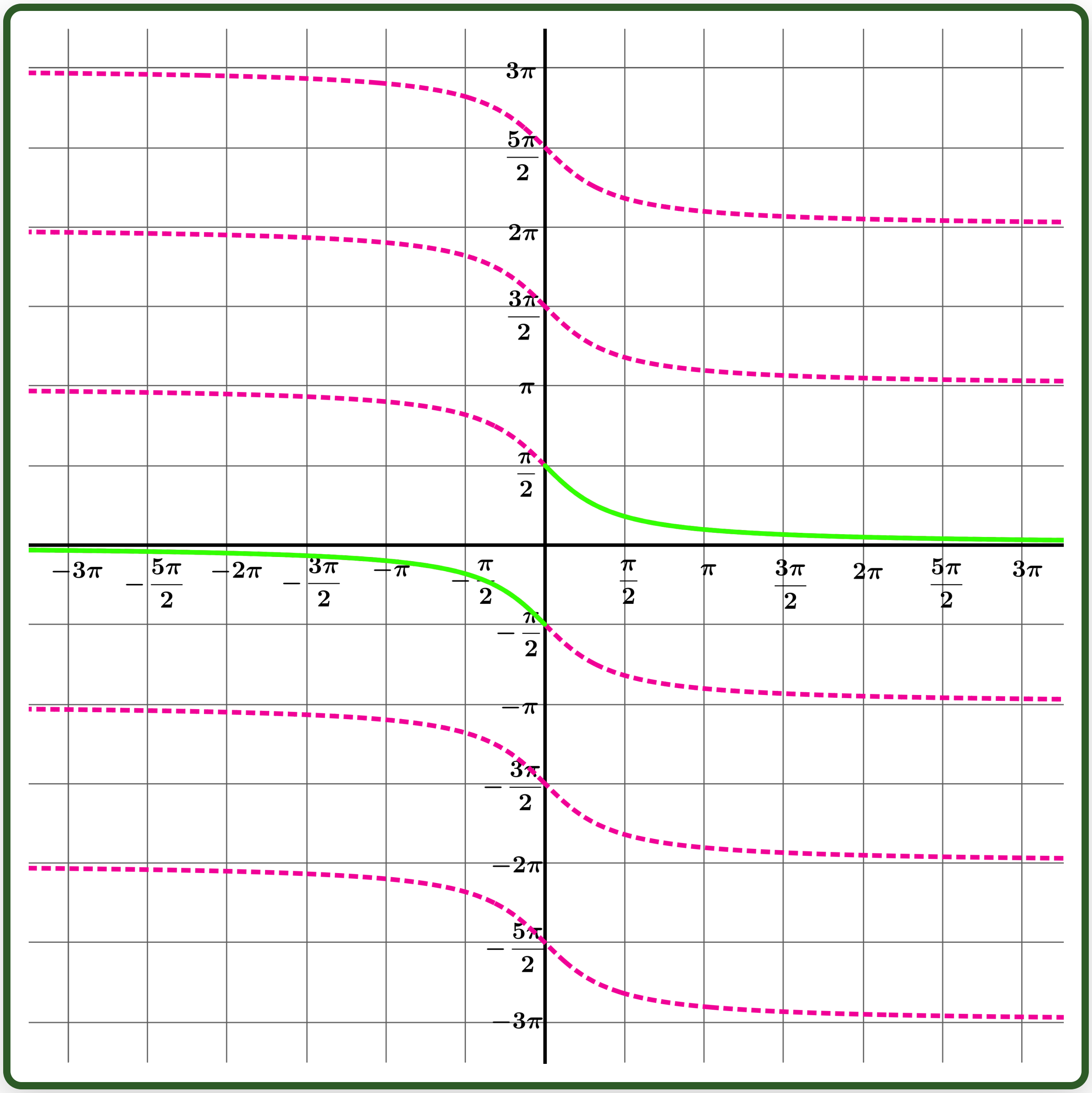
Domain: $(-\infty, -1] \cup [1, \infty)$

Range: $\left[-\frac{\pi}{2},\ \frac{\pi}{2}\right] \setminus \{0\}$

Draw the graphs $y = \cot x$ and $x = \cot y$.



Use this to draw the graph $y = \operatorname{arccot} x$

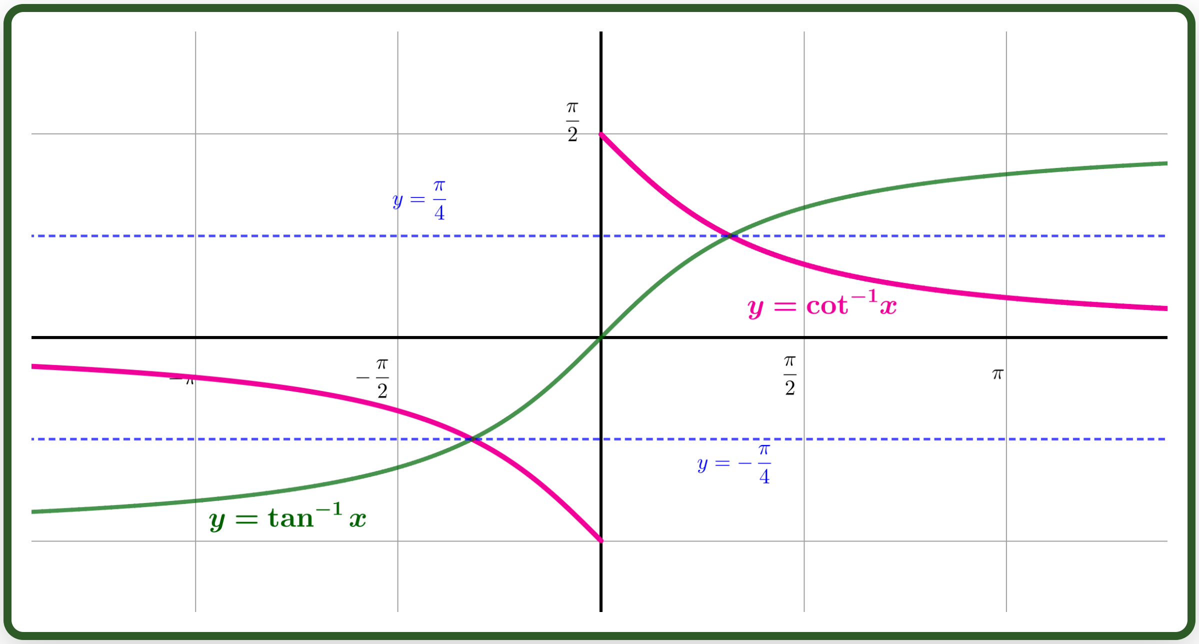


Domain: $\mathbb{R}$

Range: $[-\frac{\pi}{2}, \frac{\pi}{2}]$

The graphs of $y = \operatorname{arccot} x$ and $y = \arctan x$ are shown together below.

What symmetry do you observe? Use it to state the value of $\operatorname{arccot} x + \arctan x$.

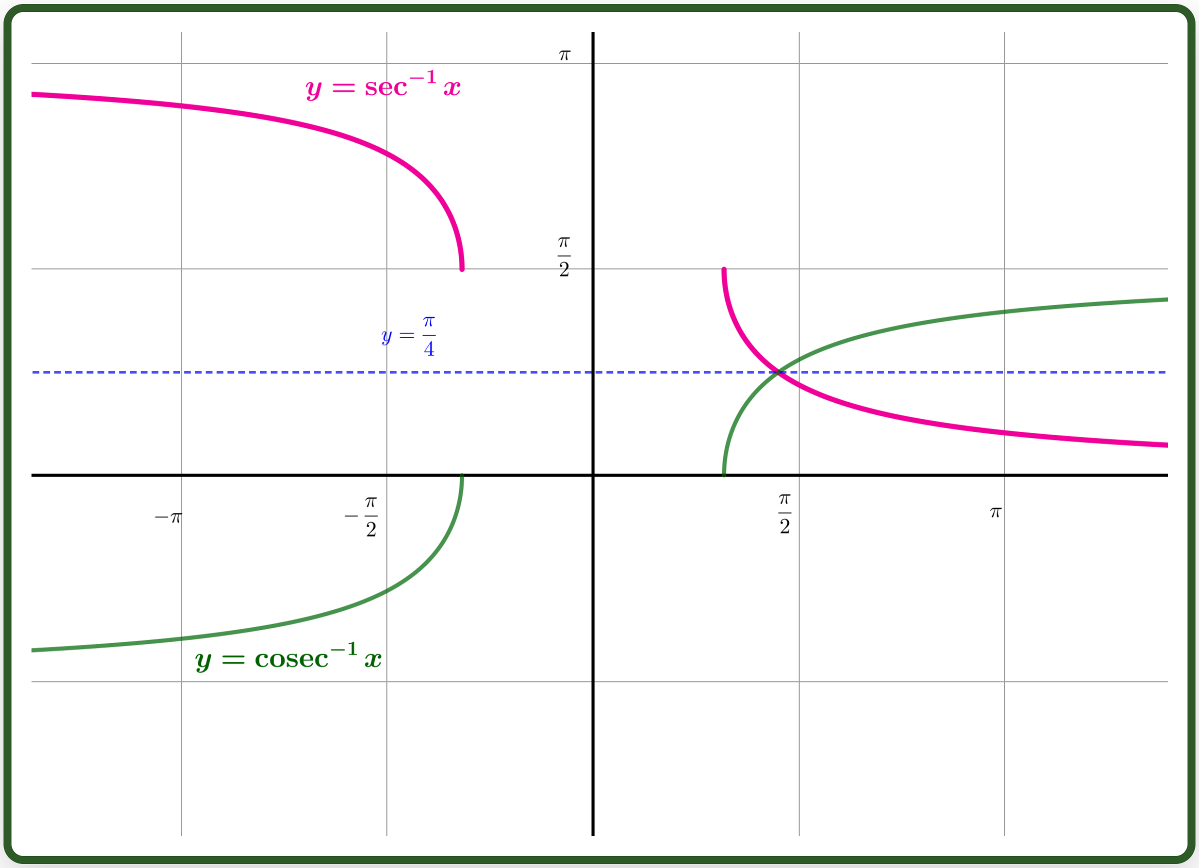


The average of $\operatorname{arccot} x$ and $\arctan x$ is either $\dfrac{\pi}{4}$ or $-\dfrac{\pi}{4}$, depending on whether x is positive or negative. So

$$\operatorname{arccot} x + \arctan x = \begin{cases} \dfrac{\pi}{2} & x > 0 \\ -\dfrac{\pi}{2} & x < 0 \end{cases}$$

The graphs of $y = \operatorname{arcsec} x$ and $y = \operatorname{arccosec} x$ are shown together below.

What symmetry do you observe? Use it to state the value of $\operatorname{arcsec} x + \operatorname{arccosec} x$.



The average of $\operatorname{arcsec} x$ and $\operatorname{arccosec} x$ is $\dfrac{\pi}{4}$. So

$$\operatorname{arcsec} x + \operatorname{arccosec} x = \frac{\pi}{2}$$

differentials and integrals of $\operatorname{arcsec} x$, $\operatorname{arccosec} x$, and $\operatorname{arccot} x$

Find $\dfrac{\mathrm{d}}{\mathrm{d}x}\operatorname{arcsec} x$

$$y = \operatorname{arcsec} x \Rightarrow x = \sec y$$

$$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}y} = \sec y \tan y = \pm \sec y \sqrt{\sec^2 y - 1}$$

$$= \pm x\sqrt{x^2 - 1}$$

$$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \pm\frac{1}{x\sqrt{x^2 - 1}}$$

but gradient on graph is always positive, so that

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{x\sqrt{x^2 - 1}}$$

Find (using integration by parts)

$$\int \operatorname{arcsec} x \,\mathrm{d}x$$

$$u = \operatorname{arcsec} x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = 1$$

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{x\sqrt{x^2-1}} \qquad v = x$$

$$\begin{aligned}\int \operatorname{arcsec} x \, \mathrm{d}x &= uv - \int v \, \mathrm{d}u \\ &= x \operatorname{arcsec} x - \int \frac{1}{\sqrt{x^2-1}} \, \mathrm{d}x\end{aligned}$$

Use the substitution $x = \sec u$ to find

$$\int \frac{1}{\sqrt{x^2-1}} \, \mathrm{d}x$$

$$\int \frac{1}{\sqrt{x^2-1}}\,\mathrm{d}x$$

$$x = \sec u \qquad \frac{\mathrm{d}x}{\mathrm{d}u} = \sec u \tan u$$

$$\begin{aligned}
\int \frac{1}{\sqrt{x^2-1}}\,\mathrm{d}x &= \int \frac{1}{\sqrt{x^2-1}}\frac{\mathrm{d}x}{\mathrm{d}u}\,\mathrm{d}u \\
&= \int \frac{1}{\sqrt{x^2-1}}\sec u \tan u\,\mathrm{d}u \\
&= \int \frac{1}{\tan u}\sec u \tan u\,\mathrm{d}u \\
&= \int \sec u\,\mathrm{d}u \\
&= \ln|\sec u + \tan u| \\
&= \ln\left|x + \frac{1}{\sqrt{x^2-1}}\right|
\end{aligned}$$

Use this result to find

$$\int \operatorname{arcsec} x\,\mathrm{d}x$$

$$\begin{aligned}
\int \operatorname{arcsec} x\,\mathrm{d}x &= x \operatorname{arcsec} x - \int \frac{1}{\sqrt{x^2-1}}\,\mathrm{d}x \\
&= x \operatorname{arcsec} x - \ln\left|x + \frac{1}{\sqrt{x^2-1}}\right| + c
\end{aligned}$$

Find $\dfrac{\mathrm{d}}{\mathrm{d}x}\operatorname{arccosec} x$.

$$y = \operatorname{arccosec} x \Rightarrow x = \operatorname{cosec} y$$

$$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}y} = -\operatorname{arccosec} y \cot y = \pm \operatorname{arccosec} y \sqrt{\operatorname{cosec}^2 y - 1}$$

$$= \pm x\sqrt{x^2 - 1}$$

$$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \pm \frac{1}{x\sqrt{x^2 - 1}}$$

but gradient on graph is always negative, so that

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{1}{x\sqrt{x^2 - 1}}$$

Find (using integration by parts)

$$\int \operatorname{arccosec} x \, \mathrm{d}x$$

$$u = \operatorname{arccosec} x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = 1$$

$$\frac{\mathrm{d}u}{\mathrm{d}x} = -\frac{1}{x\sqrt{x^2-1}} \qquad v = x$$

$$\begin{aligned}
\int \operatorname{arccosec} x \,\mathrm{d}x &= uv - \int v \,\mathrm{d}u \\
&= x \operatorname{arccosec} x + \int \frac{1}{\sqrt{x^2-1}} \,\mathrm{d}x \\
&= x \operatorname{arccosec} x + \ln\left| x + \frac{1}{\sqrt{x^2-1}} \right| + c
\end{aligned}$$

Find $\dfrac{\mathrm{d}}{\mathrm{d}x}\operatorname{arccot} x$.

$$y = \operatorname{arccot} x \Rightarrow x = \cot y$$

$$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}y} = -\operatorname{cosec}^2 y = -\left(1 + \cot^2 y\right)$$

$$= -\left(1 + x^2\right)$$

$$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{1}{1 + x^2}$$

Find (using integration by parts)

$$\int \operatorname{arccot} x \,\mathrm{d}x$$

$$u = \operatorname{arccot} x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = 1$$

$$\frac{\mathrm{d}u}{\mathrm{d}x} = -\frac{1}{1 + x^2} \qquad v = x$$

$$\int \operatorname{arccot} x \,\mathrm{d}x = uv - \int v \,\mathrm{d}u$$

$$= x \operatorname{arccot} x + \int x \frac{1}{1 + x^2} \,\mathrm{d}x$$

$$= x \operatorname{arccot} x + \frac{1}{2} \ln\left(1 + x^2\right) + c$$

Well done on completing Part 5!

You have explored the graphs, domains, ranges, and compositions of arcsec, arccosec, and arccot, and derived their derivatives using implicit differentiation.

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← Part 4